



dye synthesised or dye sensitized

Zinc Oxide Dye Synthesised Solar Cells

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Abstract

Energy demand has been on a steady increase in the latter part of the 20th century, and have reached an ever increasing growth rate of 1.8% per year. Contributing to around 0.65% of the total energy quotient, solar energy is an up and coming field whose research spectrum has been vastly expanded in the latter part of the 20th century. Solar energy, being a carbon-neutral source has both, the potential to compensate and reduce the energy obtained from the renewable sources energy sources. Fabrication of solar cells has seen many developments through the last decades but the major hindrance to 'real-time' usage has been cost and efficiency. Dye-synthesised solar cells were developed as a promising substitute to silicon cells offering clean hybrid operation, organic and inorganic materials, and low cost of installation and operation. Zinc oxide was incorporated in solar cells due to its wide band gap semiconductor nature, high electron affinity, electron diffusivity, large excitation binding energy, low-cost and stability against photo corrosion. This review sheds light on the development in ZnO dye-synthesised solar cells, synthesis of ZnO nano-structures, and composites. A variety of parameters except the choice of material and fabrication determine the type of solar cell and its efficiency such as solar radiation intensity, geographic location, cloud and shading conditions, weather conditions and module orientation. In this review, we focus on zinc oxide solar cells, their characteristics, components, working, challenges and future prospects.

Keywords: *Zinc Oxide, Dye Synthesised Solar Cells, Nano structures, Photosensitiser, Dye Regeneration, Titanium Dioxide*

Introduction

Recent studies show that the ~87% of the world energy demands are met by petroleum based sources, which are non-renewable and had adverse effects on the environment. Harnessing of solar energy is a step towards achieving an environmentally sustainable energy source. The phenomena of conversion of solar energy to electrical energy, called the photovoltaic effect was discovered by French physicist Alexandre-Edmond Becquerel in 1839, by using silver halides to synthesise solar energy. The prototype of the modern day working solar cell dates back to 1883, inventor Charles Fritts created a selenium solar cell, deposited with a thin film of gold, to create energy from photons. Although theoretically exceptional, the efficiency of this model was a mere 1%.

Silicon based solar cells were highly sought after due to their potential to achieve higher efficiency. First generation solar cells were grown on silicon wafer, mainly single crystal structures. Invented over 50 years ago, the single junction silicon solar cell, based on principle of p-n junction, by joining p- and n-type semiconductors, currently makes up more than 90% of the solar cell market. These cells employed photo generation of electrons and holes at the p-n junction, separation of the charges from the interface, which is connected via an external circuit, to generate electricity. Although having high efficiency, production costs of silicon solar cells are high. Thus, the immediate need of the hour was to engineer solar cells which provide high efficiency at lower production costs. Research in the field of nanotechnology and semiconductors paved the way for generation of thin-layered, semiconductor based, dye-synthesised photovoltaic cells, which had low production and fabrication costs. Dye-Synthesised Solar Cells (DSSCs) are a relatively new class of solar cells which are by far, the most cost-efficient medium to harness solar energy. For years, the semiconductor in use for DSSCs was titanium oxide (TiO_2). Zinc oxide (ZnO) is the best alternative to TiO_2 as both these semiconductors have a relatively close band gap, ~3.2eV and ~3.3eV, TiO_2 and ZnO respectively but ZnO has a much higher electron diffusivity and higher electron mobility ($115\text{-}155\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$) than TiO_2 which is favourable for faster and efficient transfer of electrons. Furthermore, ZnO has a large excitation binding energy of ~60eV and has experimentally shown reduced recombination reactions as compared to TiO_2 . Furthermore, ZnO is a direct band gap

semiconductor whereas TiO_2 has a more complex band structure which influences the charge separation at the surface of the anode. Since ZnO is deposited by electrochemical processes, the desired product can be tailored as per the required operation and varying the process parameters, offers a wide range of products with thickness of the ZnO film being as low as a few nanometers.

Zinc Oxide Electrodes for Dye-Synthesised Solar Cells

The third-generation solar cells, called dye-synthesised solar cells, were co-invented by Brian O'Regan and Michael Grätzel at University of California, Berkeley in 1988. The first choice for the metal oxide was Titanium dioxide (TiO_2). TiO_2 nanoparticles were used in the form of an optically transparent film, of thickness about 10 microns. The photoactive dye was used in a monolayer, to estimate the sensitivity of the cell. The overall incident photon to energy conversion efficiency was reported to be in the range of 7.1-7.9%. Practical applications of these cells was favoured by the large short circuit current density, high stability, and low costs.

Naturally occurring in the earth's crust in zincite, ZnO is an inorganic metal oxide, insoluble in water. Its micro crystals have excellent light absorbing properties in the UVA and UVB spectrum owing to its wide band gap. Having a melting point of $1,975^\circ\text{C}$, ZnO can withstand high temperature and pressure conditions. ZnO is a widely known II-VI semiconductor with less toxicity than other semiconductors, and the ability to be synthesised in large quantities at low costs. As a n-type semiconductor with a wide band gap of $\sim 3.37\text{eV}$, ZnO has excellent optical transmittance, making it a good candidate to be used in dye-synthesised solar cells. It was one of the first metal oxides used in DSSCs due to its extensive properties and a rich variety of nano structural forms, such as nano rods, nanotubes, nano walls, nano wires, with flexible synthesis and fabrication. Nanostructures of ZnO exhibit a high surface area to volume ratio, and antibacterial activity at low concentrations. The efficacy of zinc oxide in dye synthesised solar cells is determined by the efficiency of incident light absorption and electron ejection. Sensitisation on ZnO is experimentally determined to depend on the doping of the semiconductor, the immersion time of the zinc oxide electrodes in the dye, and the concentration of the dye. N-type doping of zinc with aluminium enhances carrier concentration, electrical conductivity, structural properties, and the optical absorption of ZnO. Preliminary studies were carried out using ruthenium-based dyes. It was demonstrated that longer immersion times of the ZnO electrodes in these dyes, led to a substantial decrease in the photocurrent of the solar cell. The optimum time of immersion should be such that it leads to a faster sensitisation and a rapid decrease in the photocurrent. Failing to achieve this rapid decrease, leads to accumulation of dye molecules on the electron surface causing low electron injection efficiency.

Construction of Zinc Oxide Dye-Synthesised Solar Cells

A standard dye synthesised solar cell has four components, a sensitiser (photoactive dye), a working electrode (deposited with a metal oxide in nano-form), a redox-mediator (chemical electrolyte), and a counter electrode (Figure 1)

DSSCs consists of two sheets of transparent, conductive materials which serve as a medium for deposition of semiconductor, catalyst and act as current carriers. The two essential characteristics of these substrates are a high transparency ($>80\%$) and a high magnitude of electrical conductivity. High transparency facilitates the absorption of photons by the dye and high conductivity recess energy loss and enhances electron transfer. The fluorine-doped tin oxide (FTO, $\text{SnO}_2:\text{F}$) and indium-doped tin oxide (ITO, $\text{In}_2\text{O}_3:\text{Sn}$) are usually applied as a conductive substrate in DSSCs. These substrates are essentially soda lime glass deposited with multi-layered fluorine-doped tin

oxide or indium-doped tin oxide. Indium-doped tin oxide has a higher transmittance (80%) than fluorine-doped tin oxide (75%) and a higher sheet resistance of $18 \Omega/\text{cm}^2$ compared to $8 \Omega/\text{cm}^2$ of fluorine-doped tin oxide.

The working electrode is a transparent, conducting glass plate made of ITO or FTO, deposited with a thin layer of ZnO nano-rods. The thickness of the metal oxide can range from 10-100 microns. ZnO absorbs only a minute fraction of light from the UV spectrum, so as to increase the absorbance, the electrode is immersed in a photosensitive dye. The immersion time depends on the dye concentration and nanoparticle thickness. The immersion of the electrode in the photosensitive dye leads of covalent bonding of the dye molecules on the porous surface of the metal oxide. The effective surface area of the thin film being high, large number of dye molecules are bonded which increase the rate and efficiency of photon absorption. Figure 2 and Figure 3 depict the SEM and XRD of ZnO nanorods, respectively.

The photosensitive dye ensures the maximum absorption of incident light. Initially, the dyes were ruthenium based. A good performance dye has high luminescence, strong absorption in the visible spectra, high stability at ground, oxidised and excited state, compatible redox potential, and a high efficiency of charge injection and regeneration processes. The Highest Occupied Molecular Orbital (HOMO) should be located far away from the conduction band surface of ZnO and the Lowest Unoccupied Molecular Orbital (LUMO) should be closer to ZnO. The LUMO is selected as such, that it is 0.2-0.3 eV above the conduction band of ZnO.

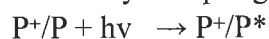
The electrolyte is carefully chosen to ensure faster reactions, less energy loss, and higher efficiency. The electrolyte has five major components, redox couple, solvents, additives, ionic liquids, and cations. The absorption spectra of the electrolyte does not overlap with that of the dye. It exhibits excellent, chemical, thermal and electrochemical stability, throughout the process. The major function of the electrolyte is to regenerate the oxidised dye. The electrolyte should not be corrosive to the glass electrodes. Acetonitrile (ACN) and N-methylpyrrolidine (NMP) are used widely due to their non corrosive nature, faster regeneration and high stability. Sometimes solvent mixtures, such as ACN/valeronitrile are used owing to their high dielectric constants.

The counter electrode is a platinum rod. It is connected to the working electrode, sealed, and electrolyte is filled. The counter electrode completes the reaction, and the ground state is restored. This facilitates the continuous movement of electrons from the working electrode, to electrolyte, to completion at the counter electrode. Platinum provides high efficiency, but has high costs. Suitable alternatives for platinum are carbon, carbon disulphide, and alloys like FeSe and CoNi.

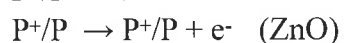
Working Principle

Working of DSSCs constitute of four major steps, absorption of photon, ejection of electron, transport of carrier and charge collection.

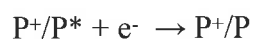
- The photons from the incident light are absorbed by the photosensitiser dye, and this absorption leads to promotion of the electron from the ground state to excited state. The energy of the photon is about 1.72 eV.
- This excited electron is injected into the conduction band of the nano-porous ZnO rods, which absorbs a fraction of the photons from the UV region, thereby oxidising the dye.
- The injected electron travels through the ZnO nanoparticles, via the electrolyte to reach the counter electrode.
- The electrons at the counter electrode are responsible for reduction, causing dye regeneration to ground state by accepting electrons from the redox mediator of the electrolyte.



Photon Absorption leading to Electron Excitation



Electron Ejection into Nanoporous ZnO



Reduction at Counter Electrode (Dye Regeneration)

The performance of a DSSC can be characterised by Overall efficiency (η), External quantum efficiency (IPCE), Fill Factor (FF), Short Circuit Current (J_{sc}) (mA.cm^{-2}), Open Circuit Voltage (V_{oc}) (volts), and Maximum Power Output (P_{max}). The plot of Current v/s Voltage represents the performance characteristics (Figure 4).

Maximum Power Output (P_{max}) is the product of Current and Voltage at maximum power

$$P_{max} = J_{mp} \times V_{mp}$$

Fill Factor (FF) is defined as the ratio of the maximum power output to the product.

$$FF = J_{mp} \times V_{mp} / J_{sc} \times V_{oc}$$

Overall efficiency (η) is the percentage of solar energy being converted to electrical energy.

$$\eta = J_{sc} \times V_{oc} \times FF / P_{in}$$

External quantum efficiency (IPCE) is defined as the ratio of number of electrons flowing in the external circuit to the number of incident photons on the cell at a given wavelength.

$$IPCE = 1240 \times J_{sc} / P_{in} \cdot \lambda$$

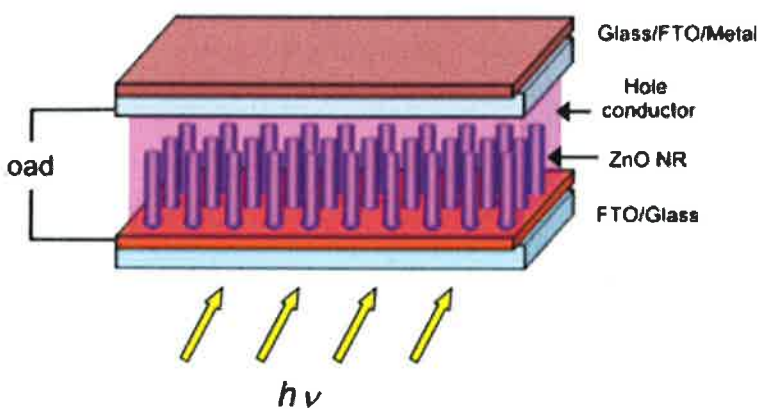


Figure 1: Dye Synthesised Solar Cell



Figure 2: SEM of ZnO Nanorods

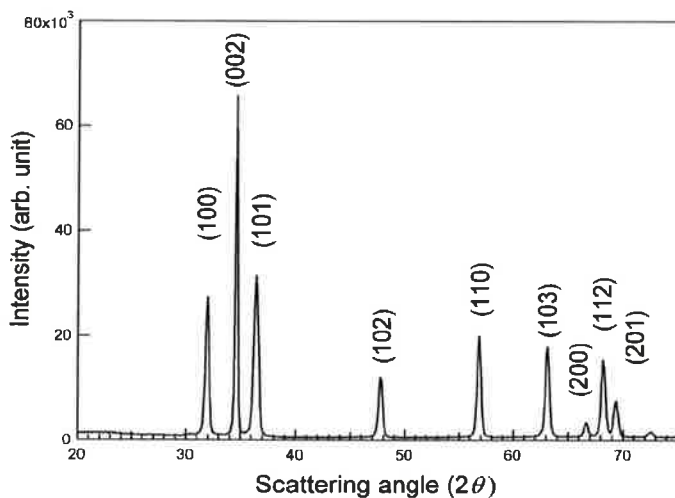


Figure 3: XRD of ZnO Nanorods

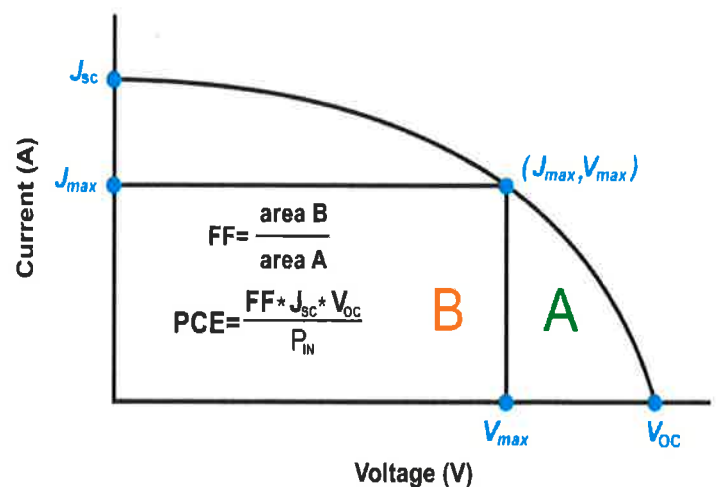


Figure 4: Performance characteristics of Dye synthesised solar cells

Challenges of Zinc Oxide Dye Synthesised Solar Cells

Even though having the benefits of easier fabrication, flexibility of synthesis and sensitisation, and low costs, the highest achieved solar energy to electrical energy efficient in ZnO-based DSSCs was reported to be 7.5%, which is considerably lower than that of TiO₂-based DSSC which is reported to be 12.3%. Stability failure, non-uniform sensitisation, and energy losses are the major challenges facing this generation of cells. Even though cost efficient than TiO₂ and the first generation silicon based solar cells, these ZnO based cells are not suitable for mass production due to their low efficiency and commercialisation obstacles.

A major reason for performance degradation is the non-uniform sensitisation of the working electrode. If the entire surface of the electrode is not uniformly covered by the sensitizer, then the spots which are not uniformly sensitised, can be occupied by the electrolyte molecules, thereby reducing the efficiency. This causes trapping of nonporous ZnO particles.

Using liquid electrolytes has major challenges. At low temperatures, the liquid can freeze causing the entire operation to cease. At elevated temperatures, liquid electrolytes are prone to expansion, which affects the selfing efficiency of the cell.

A fair amount of energy loss occurs during the regeneration process from the oxidised dye due to the energy mismatch between the oxidised dye and the electrolyte.

Stability of DSSCs can be divided into extrinsic stability and intrinsic stability. Extrinsic stability failure occurs due to the sealing materials used in the cells. These materials have low production costs and therefore are incorporated to make the process economically viable but the sealing capability of the materials decreases with increase in internal pressure. Also, subjecting the cell to cyclic or regular temperature variation degrades the sealing capability. Optimising of the sealants has still not been achieved for large module areas, thus being a major hindrance to mass production. Intrinsic stability of the cell was calculated using accelerated aging experiments. The cell was functional for 1000 hours to determine the thermal stability, dye stability, and stability of the counter electrode at a temperature of 80°C (which can easily be achieved on a regular, sunny day). Experiments showed that the cell reported an overall efficiency of 7.5-8% for a short period of time. If maintained at 50-65 °C for the given 1000 hours, the cells displayed optimum performance characteristics. But when both the factors were combined, operation for 1000 hours and at 80 °C, it was seen that the efficiency of the cell undergoes a rapid degradation. Dye synthesised solar cells need to be both, extrinsically and intrinsically stable so as to meet the market demands, and be feasible for commercialisation and mass production, as compared to silicon-based solar cells.

Methods to Overcome the Challenges

A set of various methods have been tried and tested to improve the efficiency and other performance characteristics of dye synthesised solar cells without hampering the production costs.

Incorporation of a mixture of dyes was introduced to ensure more absorption efficiency. If two or more dyes having different absorptions spectrums are used together, the absorption efficiency of the cell is enhanced, due to absorption from a broader spectrum.

Ensuring the uniform sensitisation of the working electrode prevents trapping of the nanoporous ZnO particles, and proper functioning of the cell.

The absorption efficiency can be further increased by increasing the porosity of the nonporous ZnO rods. This ensures complete absorption of the photon and prevents any energy losses off the surface.

The time taken for process of regeneration should be comparatively faster than the oxidation time of the dye. If the regeneration is not fast enough, complete restoration of the dye to the ground state

after electron regeneration does not take place, causing energy loss and incomplete energy transfer, thereby degrading the efficiency.

By incorporation novel dyes as a substitute to ruthenium-based dye. Over the years, use of natural dyes obtained from chlorophyll, anthocyanin, and beta-carotene have been used and the performance characteristics have shown a considerable improvement than those in ruthenium-based dyes. Out of these three, chlorophyll based dyes have the highest efficiency. Despite the benefits, mass production and commercialisation of these natural dye based solar cells is still in the preliminary stages of implementation.

Scope and Future Prospects

An interesting development in this field was the use of composite ZnO/TiO₂ nanoparticles instead of single-semiconductor ZnO. Experiments showed that when used in a 90/10 ratio (ZnO/TiO₂), the cells produced an efficiency similar to that of single-semiconductor cells, but a relatively higher stability. Figure 5 shows the XRD of composite ZnO/TiO₂ nanoparticles. It was also observed that increasing the ratio of TiO₂ led to a degradation in efficiency and performance characteristics. Figure 6 shows the performance characteristics graph for single-semiconductors ZnO and TiO₂, and ZnO/TiO₂ ratios 95/5, 90/10, 80/20 and TiO₂/ZnO ratios 95/5, 90/10, 80/20. It is clearly seen that a ZnO/TiO₂ nanoparticle composite has the highest efficiency while composites with a higher TiO₂ ratio have poor performance characteristics.

Implementation of multicomponent electrolyte solutions instead of liquid electrolytes can further substantially increase the efficiency. Organic electrolytes like ethanol, diethyl carbonate, dimethyl carbonate, valeronitrile, glutaronitrile, ethylene carbonate, propylene carbonate, used in suitable ratios and concentrations enhance the performance characteristics. These have high boiling points (>100°C) and low melting points(<20°C), low viscosity, low absorption, non-toxic, less soluble to sealants, and non-corrosive and available at low costs. Using quasi-solid electrolytes provide the benefits of liquid electrolytes with an added advantage of prevention from leakage, volatilization of solvent, photodegradation and desorption of dye, non-corrosive behaviour, and effective sealing for long-term use. Quasi solid state electrolytes are classified into four types: thermoplastic polymer electrolytes, thermosetting polymer electrolytes, composite polymer electrolytes, and ionic liquid electrolytes.

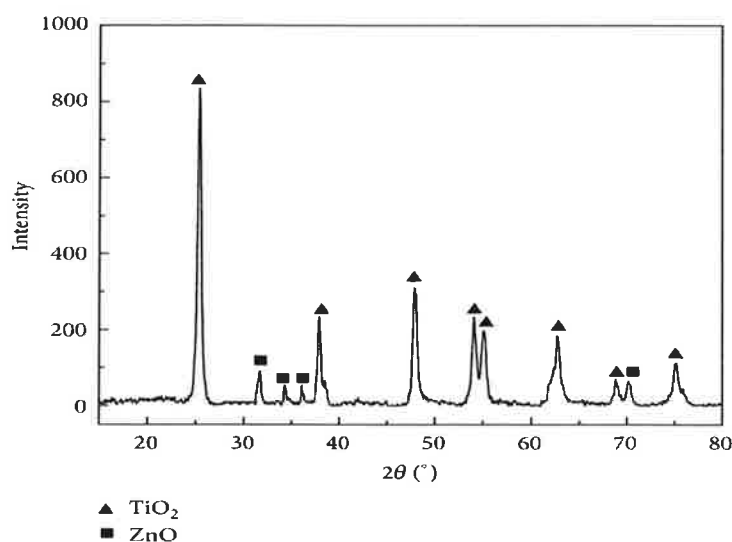


Figure 5: XRD Pattern of composite ZnO/TiO₂ nanoparticles

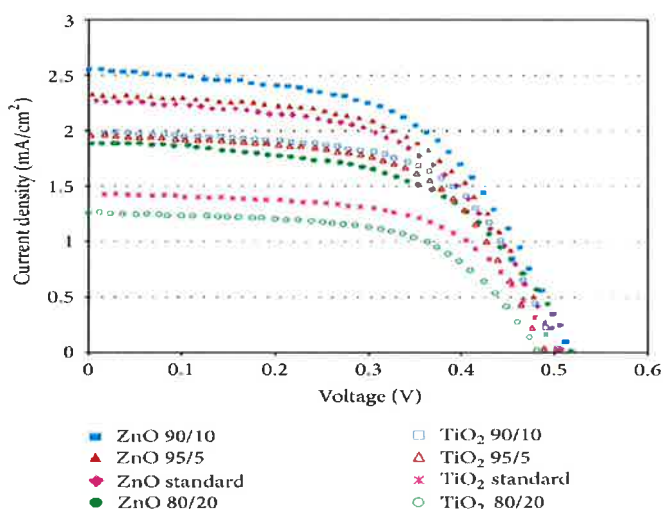


Figure 6: Performance Characteristics of single-semiconductors and composites

Conclusion

The main aim of this study was to compile a comprehensive review of Dye Synthesised Solar Cells using single-semiconductor ZnO nanostructures. Despite providing a low cost, flexible and environmentally friendly alternative to the first generation silicon-based solar cells, the major hindrance to dye synthesised solar cells still remains to be mass production and commercialisation. ZnO, having the flexibility of being used in a variety to morphological forms, and nanostructures, can be used as per the need in the form of nanorods, nanowires, nanotubes, thin-film, provides an easier operation than its counterpart, TiO₂. However convenient the process, it fails to match the efficiency and performance characteristics offered by single semiconductor TiO₂ cells. A promising solution to this was incorporation of ZnO/TiO₂ composites in a 90/10 ratio, respectively, which provides richer characteristics. A new development to this research is the use of nuclear fission to be the source of light, as an alternative to the sun. This can be highly beneficial as it eliminates the weather parameters, uncertainty and can be controlled and simulated as per the requirement. The downside to this is the high costs, which were the initial reason for use to DSSCs instead of silicon based solar cells. One of the major reasons why the silicon based solar cells happen to have a monopoly in the market is due to their high lifespan ranging over more than 25 years. For commercial application, the DSSCs should have a stable operation for a lifespan of >25 years to be a feasible replacement of silicon based cells. A comprehensive research needs to be carried out for new composite semiconductors, optimising the electrolyte parameters, dye concentration and introduction of novel, low cost dyes, with the aim of enhancing the efficiency, performance characteristics, and lifespan for commercial applications.

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Organization	19
Introduction	20
Results & Discussion	20
Conclusion / Outlook	20
Total	(79)